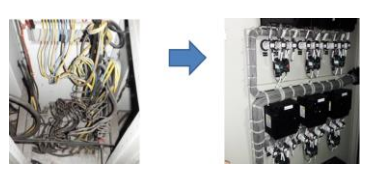
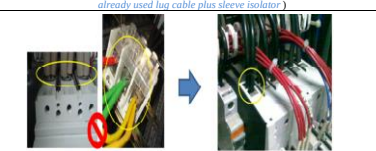
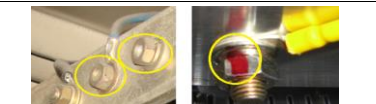


|                                                                                                     |                                      |                     |             |
|-----------------------------------------------------------------------------------------------------|--------------------------------------|---------------------|-------------|
| <br>PT. MNA SERANG | <b>WORKING INSTRUCTION</b>           | WUMNASRG-E&I-19-015 |             |
|                                                                                                     | <b>PANEL CLEANING AND INSPECTION</b> | Dept.               | ENGINEERING |
|                                                                                                     |                                      | Section             | E&I         |
|                                                                                                     |                                      | Revision            | 0           |
|                                                                                                     |                                      | Effective Date      |             |
|                                                                                                     |                                      | Page                | 1 of 1      |

| Instruksi / Instruction                                                                                                                                                                                           |                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pre-Action /                                                                                                                                                                                                      | Pre-Action                                                                                                                                                                                                                          |
|                                                                                                                                  |                                                                                                                                                    |
| Siapkan dokumen ceklis, peralatan pembersih, penggaris dan spidol marker (Prepare checklist document, cleaning tools, ruler, marker and LOTO )                                                                    | Koordinasi dengan tim produksi (Coordinate with the production team )                                                                                                                                                               |
|                                                                                                                                  |                                                                                                                                                    |
| Siapkan JSA, Work Permit LOTO dan form LOTO (Prepare JSA, LOTO Work Permit and LOTO form )                                                                                                                        | Matikan sumber energi terkait dan lakukan pemasangan LOTO pada breaker dan key switch                                                                                                                                               |
| Pembersihan dan Inspeksi / Cleaning and Inspection                                                                                                                                                                |                                                                                                                                                                                                                                     |
|                                                                                                                                  |                                                                                                                                                    |
| Bersihkan komponen-komponen dan bagian-bagian dalam panel menggunakan vacuum cleaner, sikat, dan kain lap (Clean all components and parts of the panel by using vacuum cleaner, brush and dust-cloth )            | Rapikan jalur kabel dengan kabel tie dan tutup dengan kabel ducting. (Tidy the cable strewn by using cable tie and cover with cable duct.)                                                                                          |
|                                                                                                                                 |                                                                                                                                                   |
| Periksa secara visual kondisi panel apakah ada retak, goresan atau cacat fisik. (Check panel roof if there are cracks, corrosion or damaged.)                                                                     | Periksa atap panel apakah ada kebocoran, korosi ataupun rusak (Check panel roof if there are cracks, corrosion or damaged)                                                                                                          |
|                                                                                                                                |                                                                                                                                                  |
| Pastikan base atau plat bawah sudah terpasang dan sudah disegel dengan baik (Make sure base or bottom plate is already installed and well-sealed)                                                                 | Periksa pintu panel apakah sudah terpasang PE grounding, kunci dan penahan pintu. (Check panel door if already installed by PE grounding, key and door holder.)                                                                     |
|                                                                                                                                |                                                                                                                                                  |
| Periksa ventilasi pada lubangny apakah ada cacat fisik, korosi, tertutup material asing atau ukuran terlalu kecil. (Check the hole of ventilation if there are no physical damaged or too small size.)            | Periksa terminasi kabel apakah sudah tertutup secara sempurna dan sudah menggunakan kabel lug ditambah isolator sleeve (Check the cable termination if already installed perfectly and already used lug cable plus sleeve isolator) |
|                                                                                                                                |                                                                                                                                                  |
| Periksa semua kabel sudah dilabeli sesuai dengan wiring diagram. (Check the cables if appropriate already labelled with wiring diagram)                                                                           | Pastikan terminasi kabel pada terminal blok sesuai dengan jumlah kabel (Make sure the cables termination on block terminal are appropriate with amount of cables)                                                                   |
|                                                                                                                                |                                                                                                                                                  |
| Periksa jarak terminasi pada kabel-kabel fasa pada breaker. Pastikan jarak terminasi minimal 20 mm. (Check distance of cable termination between phase to phase on breaker. Make sure the distance minimum 20 mm) | Periksa kekencangan koneksi baut dan tandai baut yang sudah kencang dengan spidol marker. (Check tightening of bolts connection and mark the tightened bolts with marker)                                                           |